Work Plan

DTSC 3601: Data Science Studio 3, Predictive Analysis and Their Impacts

SecondChance: shelter adoption forecasting for the Austin Animal Center

Davis Higgins · Submitted Tuesday, August 25, 2026

1. Partners

Davis Higgins, working solo.

I considered forming a group and decided against it for a specific reason. The project I want to build is a full serverless lifecycle: ingestion, warehouse, training, an API, a monitoring loop and a front end. In a four person team that splits into four people who each understand a quarter of it. Running it alone means every architectural decision, every leakage check and every deployment step passes through one set of hands, and I will be able to answer a question about any part of it in the final presentation without saying "that was someone else's piece." The cost is volume of work, which I have accounted for in the schedule in section 4 by cutting scope in two named places rather than by hoping to work faster.

If the workload proves unmanageable by the internal early draft deadline in Week 4, the fallback is described in section 5.

2. Direction

The theme: applying survival analysis and gradient boosting to operational decisions in animal welfare, with the monitoring and governance apparatus that a prediction affecting a living animal actually requires.

The problem. Municipal animal shelters run at capacity. Staff time, photography, foster recruitment, behaviour support and adoption event slots are all scarce, and they are currently allocated by intuition and by whoever happens to be noticed. Some animals sit for months while nobody realises they are sitting. The Austin Animal Center, the largest no-kill municipal shelter in the United States, publishes every intake and every outcome it records, which makes this a rare case where an operational allocation problem has a public, complete, long-running dataset attached to it.

What SecondChance does. For every animal, at the moment it arrives, it predicts:

- the probability of adoption within 7, 14 and 30 days
- the probability of still being in the shelter after 60 days, which is the shelter's own definition of a long-stay resident
- the full curve of adoption probability over time, and an expected number of days in care

Those predictions drive a ranked "needs attention" queue, a 30-day population forecast for capacity planning, and an outreach planner that turns the queue into a week of assignable work.

Why survival analysis and not just classification. Two thirds of the technical interest in this project lives here. An animal still in the shelter on the day the data was exported has not "failed to be adopted"; we simply do not know yet. That is right censoring, and treating it as a negative label corrupts the training set. Worse, an animal transferred to a rescue partner on day three can never be adopted from Austin afterwards, so a transfer is a *competing risk*, not ordinary censoring. Kaplan-Meier applied naively to this data will overstate adoption, and I expect that overstatement to be large enough to be worth measuring and reporting as a result in its own right.

The methods I intend to use. Cause-specific Cox proportional hazards and Aalen-Johansen cumulative incidence for the survival side; gradient boosted trees for the binary horizons; isotonic calibration on a held-out block; SHAP and permutation importance for interpretability; population stability index and Kolmogorov-Smirnov for input drift; and a language model called through Amazon Bedrock to turn free text breed strings into structured attributes.

The ethical constraint, stated up front because it shapes the design. This model must never be used to justify euthanasia, intake refusal, or reducing the care an animal receives. A low predicted adoption probability is a request for more help, not less. This is not a disclaimer I intend to add at the end. It determines what the API returns, what the interface can display, and what goes in the model card.

What I am most interested in learning. How a prediction pipeline behaves when the world underneath it changes. The Austin data spans March 2020, when the shelter's operating model changed abruptly, and I want to see what a model trained on 2018 does when it meets 2020, whether standard drift detection catches it, and how much retraining actually recovers.

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3. Sources

3.1 Data sources and APIs

Austin Animal Center Intakes. City of Austin Open Data Portal, Socrata dataset `wter-evkm`. One row per arrival, from 1 October 2013 to the present: animal ID, timestamp, found location, intake type, intake condition, species, sex, age, breed and colour. Roughly 137,000 rows as of March 2022. `https://data.austintexas.gov/resource/wter-evkm.json`

Austin Animal Center Outcomes. Same portal, dataset `9t4d-g238`. One row per departure over the same period: outcome type, outcome subtype, date of birth, species, sex and age at outcome. Roughly 137,000 rows. `https://data.austintexas.gov/resource/9t4d-g238.json`

These two are the heart of the project. They share an animal ID, which means the intake and outcome events can be paired into a real episode with a real duration label, rather than a duration inferred from a proxy. Both are available through the Socrata API with SoQL filtering, which supports incremental pulls. An application token is free and lifts the rate limit; the API works without one at a lower ceiling.

National Weather Service API. `https://api.weather.gov`. Hourly observations for the Austin station, aggregated to daily maximum, minimum and precipitation. No key required, but the API rejects requests that do not send a descriptive User-Agent header. Used to test whether weather around the intake date affects adoption speed, which is plausible because adoption is a walk-in activity. Documentation: `https://www.weather.gov/documentation/services-web-api`

RescueGroups.org Adoptable Pet API. `https://api.rescuegroups.org/v5/public`. Current adoptable listings across partner organisations, filterable by radius from a postal code. Provides regional supply context, which matters: a shepherd mix is harder to place in a week when forty other Austin-area organisations are also listing shepherd mixes. Requires a free API key from their developer portal. Documentation: `https://rescuegroups.org/services/adoptable-pet-data-api/`

Amazon Bedrock. Not a data source but an API in the pipeline. Used to convert Austin's 2,736 distinct free-text breed strings into structured attributes.

3.2 Literature I have already read

These are the papers the project is built on. The full literature review in Week 2 will expand and synthesise them; this is what I have found so far and why each one matters.

1. **Bradley, J. and Rajendran, S. (2021).** "Increasing adoption rates at animal shelters: a two-phase approach to predict length of stay and optimal shelter allocation." *BMC Veterinary Research* 17, article 70. DOI: 10.1186/s12917-020-02728-2. The closest existing work: 114,256 records from shelters in five states, comparing logistic regression, neural networks, random forest and gradient boosting. Gradient boosting won at roughly 60 percent across precision, recall and F1. The gap I intend to fill is that they treat length of stay as a classification problem and do not handle censoring or competing risks at all.

2. **Protopopova, A., Mehrkam, L. R., Boggess, M. M. and Wynne, C. D. L. (2014).** "In-Kennel Behavior Predicts Length of Stay in Shelter Dogs." *PLOS ONE* 9(12): e114319. DOI: 10.1371/journal.pone.0114319. 289 dogs, 1,766 video observations. Specific in-kennel behaviours predicted length of stay: leaning on the enclosure wall added about 30 days. Establishes that length of stay is predictable from observable features, and that morphology (small, long coat, lap dog breeds) shortens stays independently of behaviour. Their behavioural variables are exactly what Austin's public export does not contain, which bounds how well my model can possibly do.

3. **Lepper, M., Kass, P. H. and Hart, L. A. (2002).** "Prediction of adoption versus euthanasia among dogs and cats in a California animal shelter." *Journal of Applied Animal Welfare Science* 5(1), 29-42. DOI: 10.1207/S15327604JAWS0501_3. 4,813 cats and 3,301 dogs. Age, sex, coat colour, reason for

relinquishment, breed and purebred status all predicted adoption. The foundational paper for the feature set, and a useful ethical contrast: it frames the problem as adoption versus euthanasia, which is the framing I am deliberately refusing.

4. **Olson, K. R., Levy, J. K., Norby, B., Crandall, M. M., Broadhurst, J. E., Jacks, S., Barton, R. C. and Zimmerman, M. S. (2015).** "Inconsistent identification of pit bull-type dogs by shelter staff." *The Veterinary Journal* 206, 197-202. 120 dogs, 16 raters, visual identification compared against DNA. Median agreement with DNA was 67 to 78 percent and average sensitivity for identifying pit bull-type dogs was 50 percent. This is critical for me: the breed field I am about to build features from is a staff guess, and this paper quantifies how unreliable that guess is. Any breed-related finding has to be reported as a finding about the *label*, not about the animal's genetics.

5. **Sinski, J., Carini, R. M. and Weber, J. D. (2016).** "Putting (Big) Black Dog Syndrome to the Test: Evidence from a Large Metropolitan Shelter." *Anthrozoös* 29(4). DOI: 10.1080/08927936.2016.1228769. Tests the widespread shelter belief that black dogs are adopted more slowly and does not find support for it. I intend to test the same thing in the Austin data, because a model that silently encoded the folk belief would help make it true.

6. **Mavrovouniotis, M. L. (2026).** "Use of Kaplan-Meier and Cox regressions in the distribution of length of stay in animal shelters for pre-specified calendar periods." *PLOS ONE* 21(1): e0342102. DOI: 10.1371/journal.pone.0342102. Argues that the standard way shelters compute length of stay, averaging over animals whose stay ended inside a reporting window, is systematically misleading, and proposes a Kaplan-Meier and Cox formulation with left truncation and right censoring instead. Directly validates the methodological choice at the centre of my project, with worked examples from Orange County showing mean length of stay rising from 15.1 to 20.8 days across COVID.

7. **Austin, P. C., Lee, D. S. and Fine, J. P. (2016).** "Introduction to the Analysis of Survival Data in the Presence of Competing Risks." *Circulation*. The methodological reference for the competing risks treatment. Explains why the cause-specific hazard and the cumulative incidence function answer different questions and why conflating them overstates the event of interest.

8. **Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F. and Dennison, D. (2015).** "Hidden Technical Debt in Machine Learning Systems." *Advances in Neural Information Processing Systems 28*, 2503-2511. The argument that the model is a small part of a real machine learning system, and that the surrounding infrastructure is where the maintenance cost lives. This is effectively the thesis statement of the course's final product requirement, and it shapes how I am scoping the build.

9. **Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D. and Gebru, T. (2019).** "Model Cards for Model Reporting." *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. DOI: 10.1145/3287560.3287596. The format I will use for the governance deliverable, including the subgroup-disaggregated evaluation and the explicit statement of out-of-scope uses.

10. **Obermeyer, Z., Powers, B., Vogeli, C. and Mullainathan, S. (2019).** "Dissecting racial bias in an algorithm used to manage the health of populations." *Science* 366, 447-453. DOI: 10.1126/science.aax2342. A prediction system that was accurate at what it measured and harmful because of what it measured. The cautionary case for why I am checking subgroup performance by breed label rather than assuming aggregate accuracy is enough.

3.3 Technical references

- **Graf, E., Schmoor, C., Sauerbrei, W. and Schumacher, M. (1999).** "Assessment and comparison of prognostic classification schemes for survival data." *Statistics in Medicine* 18(17-18), 2529-2545. The inverse probability of censoring weighted Brier score I will use to evaluate the survival model.
- **Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. (2017).** "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." *Advances in Neural Information Processing Systems 30*.
- **Chen, T. and Guestrin, C. (2016).** "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785-794. DOI: 10.1145/2939672.2939785.
- **Lundberg, S. M. and Lee, S.-I. (2017).** "A Unified Approach to Interpreting Model Predictions." *Advances in Neural Information Processing Systems 30*.
- `lifelines` documentation for the Kaplan-Meier, Aalen-Johansen and Cox implementations: `https://lifelines.readthedocs.io`
- AWS Serverless Application Model documentation for the infrastructure as code: `https://docs.aws.amazon.com/serverless-application-model/`
- AWS documentation on Redshift `COPY` from S3, SageMaker training jobs and endpoints, and CloudWatch custom metrics and alarms.

3.4 Where I looked and did not find what I needed

Worth recording so I do not search the same ground twice. I looked for shelter datasets with behavioural assessments attached, because Protopopova et al. show that is where the remaining signal is. Sonoma County, Dallas and Long Beach all publish outcome data but none of them publish behavioural evaluations, and the Austin export does not either. I also looked at the Kaggle "Shelter Animal Outcomes" competition data and found it is a subset of the same Austin feed with fewer columns and a shorter window, so there is no reason to prefer it over the live API.

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4. Work Schedule

The course deliverable schedule sets the hard dates. My internal plan front-loads the data engineering, because everything downstream is blocked on the episode join being correct, and a mistake there is silent.

4a. How long on the exploration phase

Eight days, 18 to 25 August, roughly 22 hours. This is Week 1 and the first half of Week 2, running in parallel with writing this document.

The exploration phase is not "look at some histograms." It is answering three questions that determine whether the project is viable at all:

1. Can intakes and outcomes actually be paired into episodes? The two feeds share an animal ID but roughly one animal in eleven appears more than once, so a naive join on ID produces a cross product. I need a sequence alignment within each animal, and I need to measure how many episodes it fails on.
2. How many animals are still in the shelter at the export cutoff, and how many have an intake with no outcome ever recorded? The first group are legitimate censored observations. The second group are missing paperwork, and mixing them would poison both the survival curves and the occupancy feature.
3. Which fields are available at intake time and which are only known at outcome? The outcomes feed carries `Sex upon Outcome` and `Age upon Outcome`. Austin neuters on the way out, so `Sex upon Outcome` is close to a direct readout of whether the animal was adopted. Any model that touches it will look superb and be worthless. Building the leakage boundary explicitly is exploration work.

Deliverable at the end of this phase: a written data audit with counts for every join outcome, and a frozen episode table.

4b. When the project outline will be ready

Draft by Thursday 27 August, finalised for submission Tuesday 1 September.

The outline is due in Week 3. I want it written after the exploration phase rather than before, because the outline commits me to a set of results sections and I would rather commit to sections I already know the data can support. By 27 August I will know the episode count, the censoring rate and the class balance for each target, which is enough to write an honest abstract and a methods section that will not need rewriting.

The literature review is due 27 August and I am writing it in parallel during Week 2, drawing on the ten papers in section 3.2 plus whatever the citation chains surface.

4c. How much time on coding

Roughly 55 hours across four weeks, 25 August to 20 September, split as:

Block	Days	Hours	What
Ingestion and warehouse	25-30 Aug	12	Socrata, NWS and RescueGroups clients with retry and paging; S3 landing zone; Redshift schema; the `COPY` path; the episode builder as production code with tests
Feature engineering	31 Aug - 3 Sept	10	Intake-time-only feature set; the Bedrock breed structuring step and its evaluation; the shelter occupancy sweep
Modelling	4-10 Sept	14	Baselines, gradient boosting on four targets, Cox, Aalen-Johansen, the discrete-time competing risk hazard model, isotonic calibration, bootstrap intervals, subgroup evaluation
Interpretability and monitoring	11-15 Sept	9	SHAP, permutation importance, partial dependence; PSI and KS drift; the monthly performance report; CloudWatch metrics and alarms; the retraining experiment
Deployment and the front end	16-20 Sept	10	SAM template, SageMaker endpoint, API Gateway, the website build

I am deliberately allocating more time to the pipeline around the model than to the model itself. Following Sculley et al., that is where the work actually is, and it is also where this course's grading weight sits.

Two named scope cuts, decided now rather than under pressure later. First, I will not build a trained population forecasting model; the 30-day capacity projection will be a seasonal baseline, clearly labelled as such in the interface. Second, the return-after-adoption prediction listed in my original concept is cut, because Austin's export does not reliably distinguish a returned adoption from a new stray intake, and I would be modelling noise.

4d. When documentation starts

Documentation starts on day one, 18 August, and is never a separate phase.

Concretely: every module gets its docstring when it is written, not afterwards. The decision log gets an entry the same day a decision is made, because the reasoning behind "why is the occupancy feature capped at 289 days" is completely gone from my memory in three weeks. The data dictionary is written alongside the feature code. The model card is drafted as soon as the first model trains, with the metrics left explicitly marked as pending until they are measured.

The pieces that genuinely are a phase:

- **1-3 September:** the AWS setup runbook, written while I am doing the setup, because that is the only time the steps are actually known.
- **11-14 September:** the paper draft, due Week 5 on 17 September.
- **21-24 September:** revision after peer review, plus the final paper.
- **25-28 September:** the presentation and rehearsal.

4e. When the initial demo is finished

Working demo by Thursday 10 September, matching the internal early-draft deadline in Week 4.

"Working" here means specific things: the data pipeline runs end to end from API to episode table; at least one model is trained and evaluated on a temporal split with real numbers; and the website renders those real numbers from a static export. It does not need the live SageMaker endpoint, and it does not need the monitoring loop. Those land by 20 September.

The reason for the early date is that a demo is the fastest way to discover that a number is wrong. A metric that looks fine in a notebook looks obviously wrong when it is rendered next to an animal's name and a photograph.

4f. Full schedule against the course deadlines

Week	Dates	Course deliverable	My work
1	Aug 18-24	Quiz 1; work plan assigned	Data exploration; API smoke tests; episode join
2	Aug 25-31	**Work plan due Aug 25; literature review due Aug 27**	Ingestion and warehouse build; outline drafted
3	Sept 1-7	**Project outline due**; midterm prep	Feature engineering; AWS setup and runbook; LLM breed step
4	Sept 8-14	Midterm; internal early draft deadline	**Demo by Sept 10**; modelling; interpretability
5	Sept 15-21	**Project draft due Sept 17**; review peer draft	Monitoring and drift; deployment; paper draft
6	Sept 22-28	**Two peer reviews due Sept 24**; finalise	Revision; front end polish; presentation build
7	Sept 29	**Final paper and presentation**	Rehearsal; submission

Working rhythm: 12 to 15 hours a week, roughly two evenings plus one long weekend block. That is realistic alongside a full course load, and it is the number I will be measuring myself against rather than an optimistic one.

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5. Risk Analysis

Seven risks, in descending order of how much they would actually hurt.

Risk 1: Silent label leakage from the outcomes feed

Likelihood: high if I am careless. Impact: fatal.

The outcomes table carries `Sex upon Outcome` and `Age upon Outcome`. Austin sterilises animals before adoption, so an animal that leaves neutered was very probably adopted. Join that column into the feature matrix and the model reports an AUC in the high nineties and predicts nothing useful, because at intake time nobody knows it.

Mitigation. A hard architectural boundary: the feature construction module only ever reads intake-side columns, enforced by an explicit allow-list rather than by remembering. Plus a shuffled-label control run: train the full model on randomly permuted labels and confirm the AUC lands at 0.5. If a leaked column exists, the control will not come back to chance. This costs one extra training run and it is the single highest value check in the project.

Risk 2: The episode join is wrong in a way that looks right

Likelihood: moderate. Impact: severe.

About one animal in eleven has multiple intakes. A join on animal ID alone produces a cross product, and a cross product still produces plausible-looking durations. The failure is invisible unless you look for it.

Mitigation. Sequence alignment within each animal rather than a key join: the k-th intake is closed by the first outcome that occurs at or after it and before the (k+1)-th intake. Every rejection is counted. I will report the number of episodes built, matched, left open, and rejected because the candidate outcome fell after the next arrival, and I will check that no episode has a negative duration and that no animal changes species between intake and outcome. If the audit numbers do not add up, I stop and fix the join before doing anything else.

Risk 3: The 2020 distribution shift breaks the model

Likelihood: certain. Impact: moderate, and it is also an opportunity.

The Austin shelter's operating model changed in March 2020. Any model trained on earlier data and tested across that boundary will degrade.

Mitigation, and reframing. I am not going to avoid this by choosing a split that hides it. I am going to put the whole of 2020 and 2021 in the test block deliberately, and treat the degradation as the monitoring case study the course asks for. A real, dated, externally caused shift is far better evidence that drift detection works than a synthetic one I inject myself. The risk that remains is that the degradation is so severe the model looks useless, in which case the honest result is that it looks useless across a regime change, and the paper says so and reports what retraining recovers.

Risk 4: AWS cost overrun

Likelihood: moderate. Impact: moderate, and personal.

An idle Redshift provisioned cluster is roughly 180 dollars a month. A SageMaker endpoint left running on an `ml.m5.xlarge` is about 170 dollars a month. Two forgotten resources over a six-week project is real money out of my own pocket.

Mitigation. Redshift Serverless at the minimum 8 RPU rather than a provisioned cluster, so idle time costs nothing. SageMaker endpoints created for demonstration and torn down the same day, with the training path using transient jobs. An AWS Budgets alert at 25 dollars and a hard one at 50. A documented teardown command in the runbook, and a calendar reminder to run it. The cost model goes in the appendix of the paper with the actual figures rather than estimates.

Risk 5: The RescueGroups API key does not arrive in time

Likelihood: moderate. Impact: low.

Their key is issued by a human process with no published turnaround.

Mitigation. The RescueGroups feed powers the public discovery page and regional supply context. Neither is load bearing for the core prediction. The ingestion Lambda already treats a missing key as "skip this run" rather than an error, and the website's discovery page falls back to the Austin data. If the key never arrives, the paper reports three data sources instead of four and explains why. I am requesting the key in Week 1 so there is maximum time for it to land.

Risk 6: The breed field is a staff guess and I am about to build features on it

Likelihood: certain. Impact: moderate, mostly to the interpretation.

Olson et al. measured 50 percent average sensitivity for visually identifying pit bull-type dogs. Whatever my model learns about "bully-type breeds" is learning about the label a staff member applied, not about the animal.

Mitigation. This one cannot be engineered away, so it gets handled honestly. Every breed-related result will be stated as a result about the recorded label. The model card will carry the Olson finding explicitly. And I will report the breed subgroup performance rather than only the aggregate, because if the model is worse for the animals carrying a contested label, that is exactly the thing a reader needs to know.

Risk 7: Scope, given that I am working alone

Likelihood: moderate. Impact: moderate.

Ingestion, warehouse, four models, survival analysis, an LLM step, monitoring, deployment, a website, and seven written deliverables is a lot for one person in six weeks.

Mitigation. The two scope cuts in section 4c are already made. Beyond those, the priority order if I fall behind is: the data pipeline and the leakage checks are non-negotiable; the survival analysis is the intellectual core and stays; the monitoring loop stays because it is the course's central requirement; the

website degrades to a static export with real numbers rather than a live API; and the RescueGroups integration is the first thing cut entirely. I will make that call at the Week 4 internal deadline based on where the demo actually is, not at Week 6 when there is no room left to recover.